

DevOps lab Program 2: Develop a Multi-Stage Dockerfile for Container Orchestration

Dockerfile

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg2
GNU nano 6.2 Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json package-lock.json ./
RUN npm install

COPY . .

RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000

CMD ["node", "dist/index.js"]
```

index.js

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg2/src
GNU nano 6.2 index.js
const express = require('express');
const app = express();
const PORT = 3000;

app.get('/', (req, res) => {
  res.send('Hello from multi-stage Docker!');
});

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

package.json

```
File Edit Selection View Go ... DevOps
EXPLORER package.json x
prg2 package.json
1 {
2   "name": "second",
3   "version": "1.0.0",
4   "description": "",
5   "main": "index.js",
6
7   "scripts": {
8     "start": "node dist/index.js",
9     "build": "mkdir -p dist && cp src/index.js dist/index.js",
10  },
11   "author": "rag",
12   "license": "MIT",
13   "dependencies": {
14     "express": "^4.18.2"
15  }
```

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg2
1RV24MC075_Pragathi_h_r@dell:~$ cd DevOps
1RV24MC075_Pragathi_h_r@dell:~/DevOps$ cd prg2
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ nano Dockerfile
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ cd src
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2/src$ nano index.js
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2/src$ cd ..
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ code ..
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ npm install express

up to date, audited 69 packages in 2s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.9.3 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ npm run build

> second@1.0.0 build
> mkdir -p dist && cp -r src/* dist/

1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ node src/index.js
Server running on port 3000
^C
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ docker build -t express-app .
[+] Building 3.5s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 426B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  2.4s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a77026543 0.0s
=> [internal] load build context                                  0.3s
=> => transferring context: 102.82kB                                0.2s
=> CACHED [builder 2/6] WORKDIR /app                               0.0s
=> CACHED [builder 3/6] COPY package.json package-lock.json ./    0.0s
=> CACHED [builder 4/6] RUN npm install                           0.0s

=> CACHED [stage-1 3/6] COPY --from=builder /app/package.json ./    0.0s
=> CACHED [stage-1 4/6] COPY --from=builder /app/package-lock.json ./ 0.0s
=> CACHED [stage-1 5/6] COPY --from=builder /app/dist ./dist        0.0s
=> CACHED [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules 0.0s
=> exporting to image                                              0.2s
=> => exporting layers                                              0.0s
=> => writing image sha256:7d6cb9d4725666c5a9c60ab1ccbfad9aa13236d2e132dec36266bea28b42650 0.1s
=> => naming to docker.io/library/express-app                     0.1s
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg2$ docker run -p 3000:3000 express-app
Server running on port 3000
```

